

Siyabonga Linda kunene

ST10484674

Group 4

PROG6212

RaceDay - Entity Relationship Diagram

Entities Overview

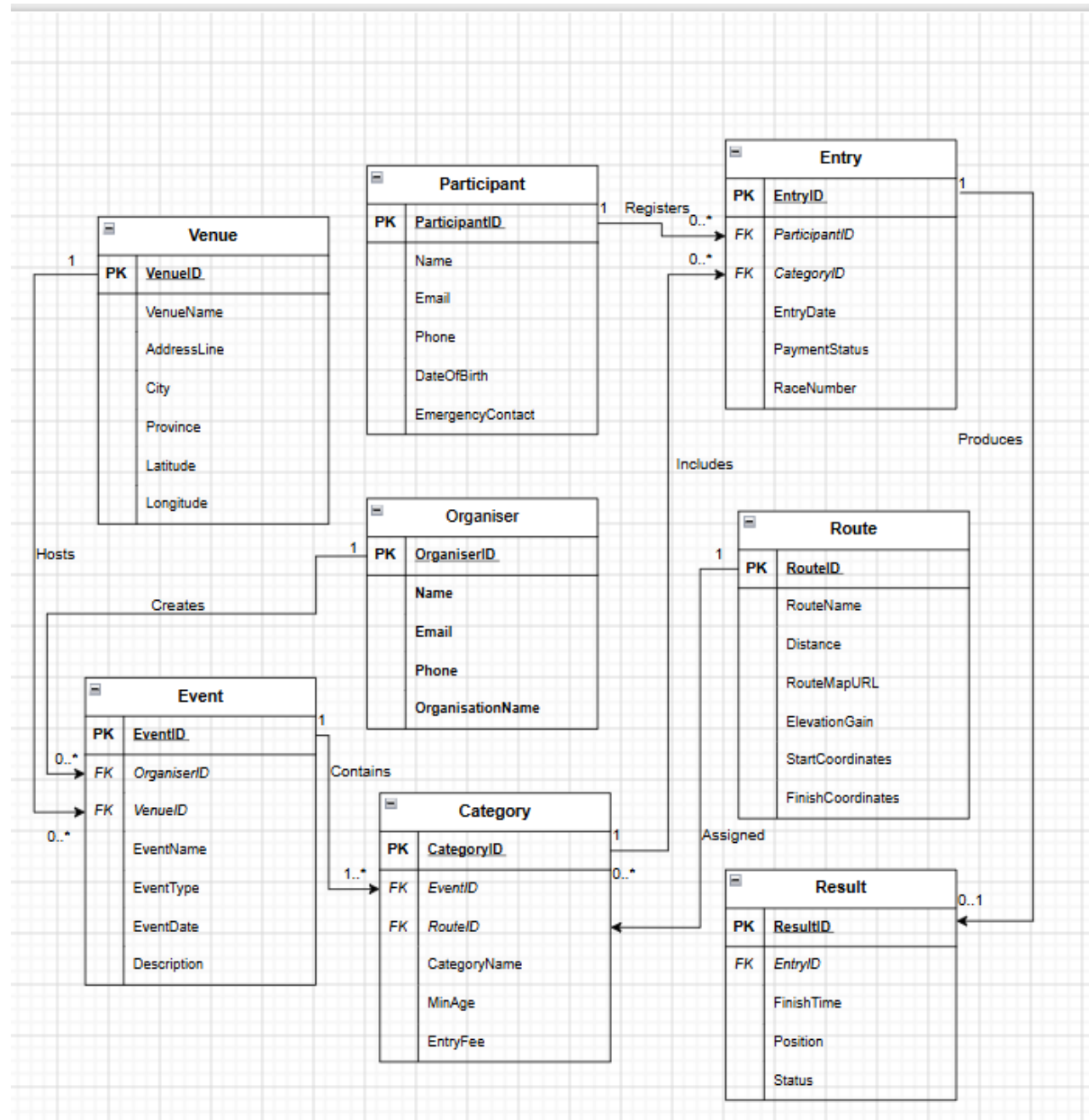
1. Organizer — a user account that creates and manages events.
2. Venue — a physical location that can host events (park, stadium, town square, etc.).
3. Event — a race day event (marathon, cycle tour, park run, etc.), held at a Venue.
4. Route — a defined course (distance, path, elevation) that a Category runs/rides/walks.
5. Category — a distance/age/type division within an event (e.g. 10km, 21km, Seniors), run over a specific Route.
6. Participant — a user account that enters events.
7. Entry — a participant's registration into a specific category of a specific event.
8. Result — the recorded outcome (time, position, status) for a completed entry.

Business Rules

1. Every Event must be created by exactly one Organizer. An Organizer may create any number of Events (1:M).
2. An Organizer may only edit or manage Events, Categories, and Results belonging to Events they personally created (enforced at the application/API layer, not the database).
3. Every Event must be held at exactly one Venue. A Venue may host any number of Events over time (1:M), enabling queries like “all events at this venue.”
4. Every Category must belong to exactly one Event. An Event may have multiple Categories but must have at least one Category before Participants can enter (1:M).

5. Every Category must be assigned exactly one Route. A Route may be reused by multiple Categories, across the same or different Events (1:M from Route to Category), so course data is never re-captured for a repeated route.
6. Every Entry must reference exactly one Participant and exactly one Category. This resolves the many-to-many relationship between Participants and Categories — a Participant may hold many Entries, and a Category may contain many Entries.
7. A Participant may not submit more than one Entry for the same Category (unique constraint on the Participant–Category pairing within Entry).
8. A Participant must meet the minimum age requirement defined on the Category (MinAge) at the time of entry.
9. An Entry cannot exist without a payment status being recorded — PaymentStatus must be set (e.g. Pending, Paid, Refunded) at the point of entry.
10. A Result may only be created for an Entry once the associated Event's date has passed.
11. Every Entry has at most one Result (0 or 1) — not all entries will necessarily result in a recorded finish (e.g. no-shows, DNF before timing capture).
12. Every Result must reference exactly one Entry — a Result cannot exist independently of an Entry (1:1, Result side mandatory).
13. Deleting an Event should be restricted (not cascaded) if it has associated Categories with existing Entries, to preserve historical participant data.
14. Deleting a Category should be restricted if Entries already exist against it.
15. Deleting a Venue or Route should be restricted if any Event or Category still references it — history must be preserved for reporting.
16. Organizer and Participant email addresses must be unique across their respective tables, since email is used for login/identification.
17. RaceNumber assigned to an Entry must be unique within the scope of a single Event.
18. Latitude/Longitude on Venue must be present if the platform is to query live weather for that location; these are required, not optional, fields.

Entity Relationship Diagram



Reference List

JGraph. 2026. *draw.io* [Diagramming software]. Available at: <https://www.drawio.com>
[Accessed: [insert the date you created the diagram]].